Marconi Infrastructure Operations Agent

An AI-Powered, Discord-Native Operations Agent for

Self-Hosted Infrastructure

Industrial Training 2026 | Marconi Lab
Implementation Phase, Weeks 5 to 8

August 2, 2026

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Abstract

This report documents the design, implementation, and evaluation of the **Marconi Infrastructure Operations Agent**, a minimum viable product (MVP) that embeds an AI-powered operations assistant inside a Discord chat interface. The agent lets infrastructure operators inspect the state of services, retrieve logs and metrics, run a bounded agentic investigation workflow driven by a self-hosted large language model (LLM), and receive background alerts when monitored endpoints degrade. It exposes the same capabilities through a FastAPI backend so that programmatic clients can reuse the workflow. The system integrates with Prometheus for metrics and Loki for logs, persists incidents in a SQLite store, and is delivered as a reproducible Docker Compose stack that also provisions a complete observability pipeline.

The agent is a two-agent system. A read-only *investigator* narrows root cause, and a separate *remediator* proposes a bounded set of write actions (Docker container restart) behind an explicit human-in-the-loop approval gate: a proposed action is persisted as pending, surfaced to an operator through Discord Approve/Reject buttons (or `/approve`/`/reject` commands), and executed only on approval, with a time-to-live expiry and a global kill switch. The LLM layer is provider-portable through a single `LLM_PROVIDER` selector that switches between local backends (Ollama, vLLM) without code changes, and a periodic container-status snapshot is posted to the status channel on a fixed cadence.

The report covers the background and motivation, the problem the agent addresses, the chosen solution and its architecture, the implementation methodology across each component, the results and operational impact observed during the internship, the challenges encountered, and the planned future work. All implementation references use the fully-qualified Python import path of the relevant module so that the document remains intelligible independently of the source repository.

1 Background

Infrastructure operations teams manage many services across different environments. The day-to-day work of an operator consists largely of situational awareness: answering *is this service healthy?*, *what do its recent logs show?*, and *what do its metrics say right now?* These questions are answered today by consulting several tools: a container runtime, a metrics store such as Prometheus, a logs store such as Loki, and an alerting pipeline. Each tool has its own query language, its own dashboard, and its own authentication boundary. The cognitive cost of context switching between these surfaces is high, and the latency between a symptom appearing and an operator gathering the relevant signals can dominate the time to mitigation.

The Marconi Lab internship programme introduces trainees to this operational reality through an implementation phase in which a small but complete system is built and demonstrated. The charter for this project was to deliver a working MVP of an AI-powered infrastructure operations agent that lives in Discord, the chat platform already used by the team, and that helps operators inspect selected systems, investigate issues through a self-hosted language model, and produce clear operational summaries. A deliberate constraint of the charter was that the agent should act through controlled, read-only tools, keeping mutation actions out of the autonomous loop until a human-in-the-loop approval mechanism exists.

Two technology trends made this project timely:

- **Self-hostable inference.** The maturity of OpenAI-compatible inference servers (Ollama [5], vLLM [6], llama.cpp, and hosted gateways) means a capable language model can be run on commodity hardware under the operator's own control, removing the data-handling concerns associated with third-party hosted APIs.
- **Standardised tool calling.** The consolidation of the `langchain-openai` [3] ecosystem around the OpenAI tool-calling protocol [2] means a language model can be given a small set of typed tools and trusted to select and invoke them correctly within a bounded loop, which is the foundational mechanism for an agentic investigation.

2 Problem Statement

The core problem is the gap between the moment an operator (or an automated monitor) suspects a degraded service and the moment a useful summary of the situation is in hand. Concretely, the MVP targets the following pain points:

1. **Fragmented signal gathering.** To answer a single question about a service, an operator must visit a container runtime, a metrics store, and a logs store, each with a distinct query language. The cost compounds when several services are involved.
2. **Manual query composition.** Even experienced operators must recall metric names, label values, and LogQL selectors before they can write a query. Discovering the available vocabulary is itself a step.
3. **Reactive, attention-driven monitoring.** Without a continuously running background check, degradations are noticed only when a human happens to look. Polling scripts exist but are typically disconnected from a notification surface the team already watches.
4. **Unstructured incident memory.** Investigations produce summaries that live in chat scrolls or personal notes, with no durable record of what was found, when it was opened, and whether it was resolved.
5. **Hosted-model constraints.** Many organisations cannot send operational telemetry to a third-party hosted LLM for compliance reasons. A solution that depends on a hosted model is therefore not universally deployable.

The MVP narrows the problem space deliberately to *read-only investigation and structured alerting*, with mutation confined to a separate, explicitly human-gated remediation agent rather than an autonomous step. This scoping is a design decision, not a limitation to be fixed: the autonomous investigation loop remains safe to run in a real environment because it cannot mutate state, and the remediation agent can only *propose* an action whose execution is gated by a human approval (with a time-to-live expiry and a global kill switch). This keeps the dangerous capability narrowly scoped while the tooling and review practices mature.

3 Solution Overview

The solution is a single Python package, `agent`, that exposes two entry points sharing one codebase: a Discord bot and a FastAPI HTTP server. Both route through the same tools, the same LLM client, and the same SQLite incident store, so that a capability added once is reachable from chat and from HTTP alike.

3.1 Design principles

The implementation follows a small set of explicit principles:

- **Read-only investigation, gated remediation.** The investigator agent is given only tools that inspect infrastructure. Mutation is confined to a separate remediation agent whose write tool (`propose_restart`) records a pending approval rather than executing; execution happens only through `execute_approval` on human approval. The raw Docker/SSH mutation functions are library-only building blocks.
- **Graceful degradation.** Every external dependency (Prometheus, Loki, the LLM) is optional at runtime. When a source is unreachable, the tool returns a stub or a fallback summary rather than raising, so the bot keeps responding during partial outages.
- **Bounded autonomy.** The agentic loop is capped at a configurable number of tool calls (`MAX_TOOL_STEPS`, default 6). If the model has not converged, a final tool-less invocation is used to force a prose summary.
- **One codebase, two surfaces.** Discord commands and HTTP routes are thin adapters over a shared workflow and tool layer, preventing drift between the chat and programmatic interfaces.
- **Self-hosted, portable, and provider-switchable.** The LLM is reached through an OpenAI-compatible base URL, so any conformant server (Ollama, vLLM, llama.cpp, or a gateway) can serve the model. A single `LLM_PROVIDER` selector switches between configured local backends without code changes. The deployment is containerised and reproducible.

3.2 System at a glance

The system comprises six logical layers, each realised as a Python subpackage under `agent`:

1. **Interface layer** (`agent.bot`, `agent.api`): Discord client, prefix and slash commands, the @mention conversational handler, the background monitor, interactive menus, a reply-summarisation layer, and the FastAPI routes.
2. **Workflow layer** (`agent.workflows`): the incident investigation loop, the remediation agent and its human-in-the-loop gate, and the reporting workflow that orchestrate the LLM and the tools.

3. **Tool layer** (`agent.tools`): controlled tools for status, logs, metrics, HTTP health, Docker, SSH, and the LangChain query tools the LLM calls, including the gated `propose_restart` tool the remediation agent uses.
4. **Source layer** (`agent.tools.sources`): thin HTTP clients for Prometheus and Loki.
5. **Storage layer** (`agent.storage`): the SQLite repository for incidents *and* pending remediation approvals, and the `Incident` data model.
6. **Support layer** (`agent.config`, `agent.llm`, `agent.alerting`): configuration, the LLM client factory, and Discord alert embed helpers.

Figure 1 shows how a request flows from either surface through the shared workflow and tool layers down to the monitoring sources and the SQLite store.

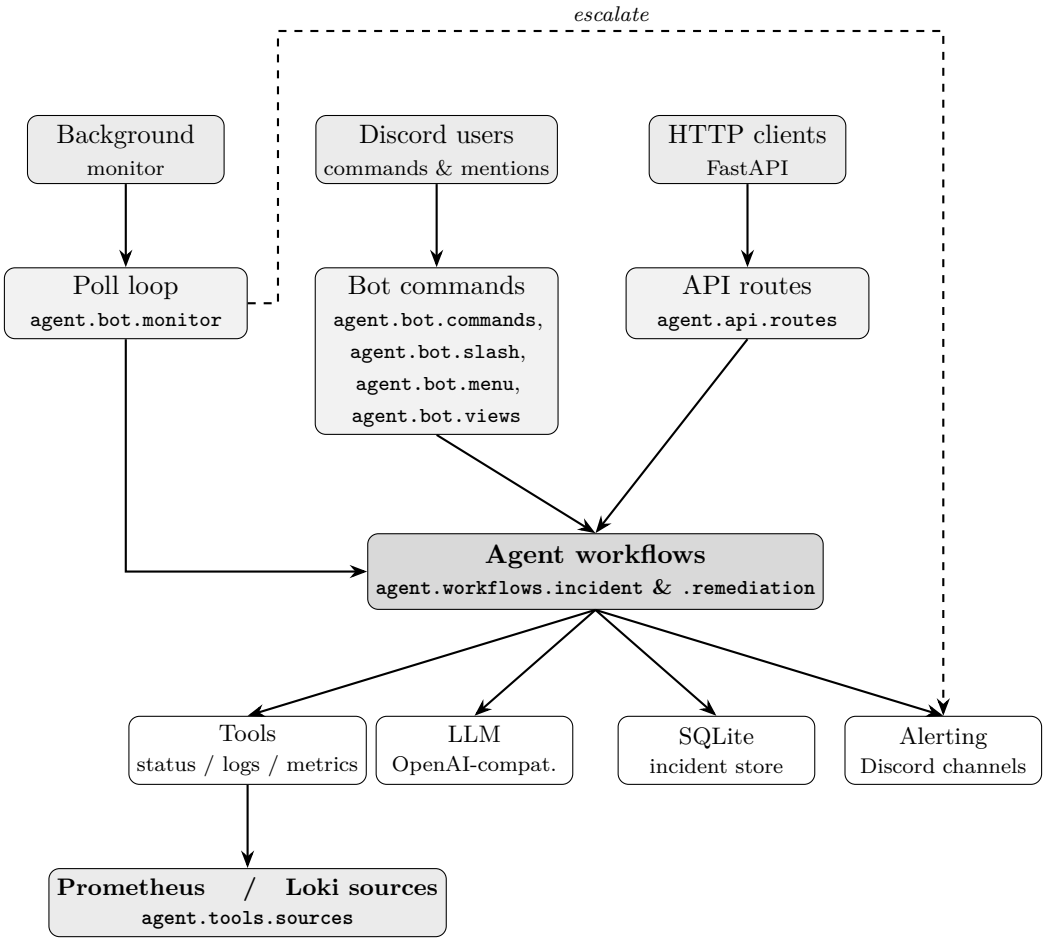


Figure 1: High-level architecture and request flow. Discord commands, HTTP requests, and the background monitor all converge on the shared agent workflow layer, which holds two agents: a read-only investigator and a gated remediator. The workflow drives read-only tools and the LLM, persists incidents and pending remediation approvals to SQLite, and posts alerts to Discord. The dashed feedback edge shows the monitor escalating a down endpoint or firing alert rule to the alerting layer, routed along the top margin to avoid crossing other boxes. The remediation agent writes only a pending approval; execution of a restart is gated by a human approval (not shown) before it reaches the Docker daemon.

4 Implementation Methodology

This section documents each component of the system, the design choices behind it, and the concrete behaviour it exhibits. The implementation is written in Python 3.13 and managed with `uv`; the runtime dependencies are `langchain-openai`, `discord.py`, `fastapi`, `uvicorn`, `pydantic-settings`, `python-dotenv`, `httpx2`, and `pyyaml`.

4.1 Configuration and settings

All runtime configuration is centralised in `agent.config.settings`, a Pydantic [10] `BaseSettings` subclass loaded from environment variables and an optional `.env` file:

- **Loading order.** A dedicated loader, `agent.config.dotenv_loader`, calls `find_dotenv` and `load_dotenv` with `override=False` before the settings object is constructed, so environment variables always take precedence over the file. This ordering matters for container deployments where secrets are injected as environment variables.
- **Field groups.** The model carries five groups of fields: Discord channel identifiers, endpoint monitoring parameters, the LLM endpoint, the API server, and the monitoring source URLs with their optional bearer tokens.
- **Empty-string normalisation.** A `@model_validator(mode="before")` normalises empty-string environment values back to empty strings for `str` fields, because Pydantic's `env_parse_none_str` would otherwise turn them into `None`, which would break downstream string concatenation.

4.2 Discord interface

The Discord client is implemented in `agent.bot.client` and built on the `discord.py` [4] library with the `commands.Bot` extension. The bot enables the Message Content privileged intent (which prefix commands depend on) and exits with a clear diagnostic if the intent is not granted or the token is invalid. Its behaviour splits across four areas, all over a shared tool and workflow layer:

- **Command surfaces.** Two parallel surfaces share one tool and workflow layer: prefix commands (`!`) in `agent.bot.commands` and slash commands (`/`) in `agent.bot.slash` through the `CommandTree`. Slash commands give operators autocomplete and a structured parameter UI; on startup the bot syncs them to the configured guild (fast, for testing) or globally.
- **Event handlers.** The `on_ready` handler is guarded against re-entrancy after a gateway reconnect by a module-level `_setup_done` flag, so the monitor and the log handler start only once. On first ready it starts the background monitor, optionally attaches a Discord log handler, syncs the slash-command tree, and probes the LLM for tool-calling support. The `on_message` handler dispatches `!` messages to the command processor and routes direct messages and @mentions through the agentic investigation loop (`run_ask_workflow`), injecting an environment context string (monitored endpoints, alert rules, channel identifiers, source URLs, and the available tool list) so answers are grounded in the actual deployment.
- **Reply handling.** Long replies are split by `agent.bot.utils.split_discord_text` to respect Discord's 2000-character limit, splitting at the nearest newline boundary. Slow slash commands defer the interaction before any network I/O, then follow up, so they never miss Discord's interaction window.
- **Interactive views.** `agent.bot.menu` presents a Discord select dropdown of container names (capped at 25 options by the platform) and, once a container is chosen, offers ephemeral Status, Logs, and Metrics buttons. `agent.bot.views.ServicePanelView` offers a service

picker with Status, Health, Logs, Metrics, and Incident buttons for the monitored endpoints. The persistent incident-action view (`agent.alerting.discord.IncidentActionView`) carries Acknowledge, Resolve, and Remediate buttons; the Remediate button runs the gated remediation agent, whose proposals are posted with their own Approve/Reject view (`RemediationApprovalView`).

Table 1 summarises the slash commands. Each has a prefix equivalent (e.g. `/status` and `!status`) for operators who type in chat.

Table 1: Discord slash commands (modules `agent.bot.slash` and `agent.bot.commands`).

Command	Description
`/status <service>`	Current status of a service via Docker container discovery
`/containers [raw]`	List Docker containers grouped by state; `raw` skips the LLM summary
`/logs <service> [n] [raw]`	Last `n` log lines from Loki (default 10); `raw` shows full logs
`/metrics <service> [raw]`	Current metrics from Prometheus; `raw` skips the LLM summary
`/health <url>`	Reachability and latency of an HTTP endpoint
`/watching`	List endpoints configured for background monitoring
`/ask <question>`	Free-form agentic investigation using read-only tools
`/incident <service>`	Run the incident workflow, open a thread, alert
`/scenario <name>`	Run a simulated scenario from a YAML file
`/ack <id>`	Acknowledge a stored incident
`/resolve <id>`	Resolve a stored incident
`/remediate <id>`	Run the gated remediation agent against an incident's diagnosis
`/approve <id>`	Approve and execute a proposed remediation action (button fallback)
`/reject <id>`	Reject a proposed remediation action (button fallback)
`/menu`	Interactive container picker with action buttons
`/panel`	Monitored-service picker with Status/Health/Logs/Metrics/Incident buttons
`/info`	Bot info and support links

4.2.1 Reply summarisation

Verbose tool output (container lists, logs, metrics) is routed through `agent.bot.summarize.clean`, a thin pass that sends the raw output to the LLM with instructions to produce a short, scannable, Discord-markdown summary. The function guards against failure at every step:

- Empty raw input returns a placeholder; an unreachable or empty LLM response falls back to the raw output in a code block; the result is capped to a Discord-safe length.
- The service argument is sanitised (newlines collapsed, length capped) before it enters the prompt, closing a prompt-injection surface.
- Each verbose command accepts a `raw` opt-out (a prefix flag or a slash option) so an operator who needs the complete, unsummarised output can bypass the LLM pass. Short, already-clean commands (`status`, `health`, `watching`) skip the summariser entirely and return raw output for low latency.
- A custom logging handler, `agent.bot.client.DiscordLogHandler`, forwards `WARNING` and above log records to a configured Discord channel. To avoid a feedback loop, it skips any record whose logger name starts with `discord`. It uses an asynchronous queue and a pump task so that `emit()` never blocks the calling thread, and it writes send failures to `stderr` rather than re-entering the logging system.

4.3 HTTP API

The FastAPI server [11] lives in `agent.api.server` and `agent.api.routes`:

- **Shared-secret auth.** The application is created with an optional shared-secret dependency: `_require_api_key` compares the `X-API-Key` header against `settings.api_key` and raises HTTP 403 on mismatch. When `API_KEY` is unset the dependency is a no-op, and the recommendation is to keep `API_HOST=127.0.0.1` so the open API is not network-reachable.
- **Routes.** The routes are thin wrappers over the same tools the Discord commands use: status, logs, metrics, HTTP health, and the incident workflow. Swagger UI is served at `/docs`.

  Table 2 lists the endpoints.

Table 2: HTTP API endpoints (module `agent.api.routes`).

Method and path	Description
`GET /health`	Liveness probe returning `{"status":"ok"}`
`GET /services/{service}/status`	Service status via Docker discovery
`GET /services/{service}/logs?lines=N`	Recent logs (default 10)
`GET /services/{service}/metrics`	Service metrics from Prometheus
`GET /health-check?url=...`	Reachability and latency of an endpoint
`POST /incidents/{service}`	Run the incident workflow, returns id and summary

4.4 Agent tools

The tool layer, `agent.tools`, is the controlled vocabulary the agent uses to inspect infrastructure. Tools are grouped by signal type and exposure. The design rule is that *only read-only tools are exposed to the investigator agent and to chat*; the Docker and SSH mutation functions are library-only building blocks, reached indirectly and safely through the gated `propose_restart` tool of the remediation agent (Section 4.7).

4.4.1 Service status

`agent.tools.services.get_service_status` matches a service name to a running Docker container via `agent.tools.docker.find_container`, which performs a case-insensitive substring match against container names. It returns a structured dict with the container name, state, status detail, and image, backed by live Docker container discovery.

4.4.2 Metrics and logs

`agent.tools.metrics.get_service_metrics` constructs a PromQL instant query for the service and delegates to the Prometheus source helper. A small `_escape_label_value` function escapes backslashes and quotes in the service name before interpolation into the query string, preventing a naive injection. On any exception it falls back to a zeroed stub dict. Similarly, `agent.tools.logs.get_recent_logs` runs a LogQL instant query against Loki and falls back to a stub log line when Loki is unreachable.

4.4.3 HTTP health

`agent.tools.health.check_endpoint` performs an HTTP GET with a 5-second timeout, follows redirects, and returns a dict with `reachable`, `status_code`, and `response_time_ms` measured with a monotonic clock. On failure it returns `reachable=False` with the error string rather than raising, which makes it safe to use directly in the monitoring loop.

4.4.4 Docker and SSH tools

`agent.tools.docker` talks to the Docker Engine API [9] over the configured `DOCKER_HOST` (a Unix socket by default, or a TCP endpoint). It exposes `list_containers`, `find_container`, `container_logs`, and `restart_container`. A `_docker_available` guard returns `False` when the socket is absent, so the list functions degrade to an empty list rather than erroring. `find_container` matches by name in a separator-agnostic way through `agent.tools.normalize`: a user spelling of `agent_api` matches the real container `marconi-agent-api` because the helper collapses `_`, `-`, spaces, and dots before comparing. `agent.tools.ssh` wraps the system `ssh` binary with `BatchMode=yes` and `StrictHostKeyChecking=accept-new` and exposes `run_ssh_command`, `check_server_load`, and `restart_service`. The latter validates the systemd unit name against the regular expression `^[A-Za-z0-9._@-]+$` to prevent command injection and requires an explicit user argument, refusing an implicit root.

4.4.5 LangChain query tools

The tools the LLM actually calls during an investigation are defined in `agent.tools.query_tools` as LangChain `@tool` functions. They split into three groups. *Discovery* tools let the model learn the real names and vocabulary of the environment before querying:

- `list_services()` returns the live Docker container names and the configured monitored-endpoint map. The system prompt tells the model to call this first and use the names it returns, not the user's spelling.
- `list_metric_names(match)` discovers available Prometheus metric names, returning at most 250.
- `get_label_values(label)` discovers values for a Prometheus label such as `service`, `job`, or `instance`.

Docker inspection tools (read-only) cover containers Prometheus and Loki do not tag:

- `docker_status(container)` returns a container's state, status, and image, resolving the name through the separator-agnostic matcher.
- `docker_logs(container, tail=50)` fetches a container's combined stdout/stderr logs via the Docker Engine API, strips stream-framing control bytes, caps `tail` at 200, and truncates to 4000 characters.

Query tools wrap the Prometheus and Loki source helpers with safety clamps:

- `query_prometheus(promql)` runs an instant query, truncating the result to 4000 characters.
- `query_prometheus_range(promql, minutes)` runs a range query with the window clamped to `RANGE_MAX_HOURS`.
- `query_loki(logql, limit)` runs a LogQL instant query with the limit clamped to `LOKI_MAX_LIMIT`.
- `query_loki_range(logql, minutes, limit)` runs a LogQL range query, clamping both window and limit.

The raw mutation functions (`restart_container`, `restart_service`, `run_ssh_command`) are not bound directly to the investigator loop; the remediation agent reaches `restart_container` only through the gated `propose_restart` tool, which records a pending approval instead of executing. The truncation and clamping are deliberate: they keep tool responses within the LLM context window and prevent runaway queries from dominating the loop. The docstrings on each tool are written as instructions to the model (*"Call this FIRST to discover what metrics exist"*), which measurably improves the model's discovery behaviour.

4.5 Monitoring sources

The source helpers live in `agent.tools.sources.prometheus` [7] and `agent.tools.sources.loki` [8]. Each is a thin `httpx.AsyncClient` wrapper with a 10-second timeout and optional bearer-token authentication drawn from the settings. The Prometheus helper implements instant and range queries, metric-name discovery, and label-value discovery; the range query computes a step size of `max(15, minutes * 60 // 100)` seconds. The Loki helper implements instant and range queries using the `/loki/api/v1/query` and `/query_range` endpoints.

4.6 Agentic investigation workflow

The investigation workflow is the intellectual core of the agent. It is implemented in `agent.workflows.incident` and consists of three entry points sharing one bounded tool-calling loop: `run_incident_workflow` (persists an incident), `run_ask_workflow` (free-form, no persistence), and the underlying `run_agentic_investigation`. The loop is parameterised on its system prompt and tool set, so specialist agents reuse the same loop with their own prompt and a narrower tool subset. The remediation agent in particular reuses it to drive its gated propose-and-approve path.

The loop is a standard ReAct-style cycle [1] with explicit bounds:

1. A system message sets the persona (*"You are Oracle, an infrastructure-operations investigator"*) and instructs the model to call `list_services` first to learn the real container and endpoint names, then discover metric names and label values, to use the actual names it discovers (not the user's spelling), to use the read-only Docker tools for containers Prometheus and Loki do not cover, and to summarise in one paragraph. Incident investigations use an extended prompt, `INCIDENT_SYSTEM_PROMPT`, that asks for two to three ranked root-cause hypotheses, each with the specific evidence (metric value, log line, or container state) that supports it, rather than a single narrative, so an operator can verify a ranked list quickly. Free-form `/ask` questions keep the general prompt, since they are questions rather than incidents.
2. A human message carries the prompt (the service name or the user's question).
3. The model is invoked with the tools bound. If it returns tool calls, each call is executed, its output is truncated to 4000 characters, and a `ToolMessage` is appended.
4. The cycle repeats, capped at `MAX_STEPS`. If the step cap is hit with pending tool calls, a final tool-less invocation forces a prose summary instead of another tool call.
5. If the model returns no tool calls, the loop ends and the content is the summary.

The workflow has a two-level fallback. If the agentic loop raises or yields an empty summary, `_static_summary` assembles a status-plus-metrics-plus-logs context and asks the LLM for a one-paragraph summary. If the LLM itself is offline, the static fallback returns the raw context string so the incident is still recorded. This means the agent degrades gracefully along three tiers: agentic, static LLM summary, raw context. Listing 1 shows the core loop.

Listing 1: Bounded tool-calling loop (module `agent.workflows.incident`).

```
async def run_agentic_investigation(
    prompt: str, *, system_prompt=SYSTEM_PROMPT, tools=TOOLS,
) -> str:
    by_name = {t.name: t for t in tools}
    llm = get_llm().bind_tools(tools)
    messages = [SystemMessage(system_prompt), HumanMessage(prompt)]
    ai = None
```

```
    for step in range(settings.max_tool_steps):
        ai = await llm.ainvoke(messages)
        messages.append(ai)
        if not ai.tool_calls:
            break
        for call in ai.tool_calls:
            name = call.get("name")
            args = call.get("args") or {}
            tool_id = call.get("id", "call")
            try:
                out = await by_name[name].ainvoke(args)
            except Exception as exc:
                out = f"tool error: {exc}"
            messages.append(ToolMessage(str(out)[:4000], tool_call_id=tool_id))
    if ai and ai.tool_calls:
        ai = await get_llm().ainvoke(messages)  # tool-less final summary
    return (ai.content if ai else "") or "No summary produced."
```

A helper, `llm_supports_tool_calls`, probes the configured model by binding the tools and issuing a task that genuinely requires one (asking it to list the running containers through `list_services`). A capable model emits a tool call; a model that answers in prose does not. The boolean is used at startup to warn the operator when the model cannot drive the agentic path. The probe deliberately uses a concrete task rather than a vague "pick a tool" prompt, which under-reported capable local models that answer such prompts in prose while still calling tools on real investigations. The `@mention` handler and `/ask` both route through `run_ask_workflow`, which is the same loop without the persistence step.

4.7 Remediation agent and human-in-the-loop approval

The remediation agent, `agent.workflows.remediation`, is the system's second agent and the only path through which a write action reaches infrastructure. It reuses the investigator's parameterised loop with three distinct parts:

- **Tool subset.** The agent runs with its own prompt (`REMEDIATION_PROMPT`) and a narrow tool set: `list_services`, `docker_status`, and a gated `propose_restart` built per run as a closure that binds the incident id, so the model cannot misroute an approval to another incident.
- **Structural gate.** `propose_restart` resolves the target container through `docker.find_container`, records a `pending` row in the `pending_approvals` table, and returns the approval id to the model; it never calls `docker.restart_container`. The only function that executes a restart is `execute_approval`, invoked on human approval. It refuses a missing or already-resolved approval, marks an approval `expired` when it is older than `REMEDIATION_APPROVAL_TTL_SECONDS` (default 300), and marks it `failed` if the restart raises. `reject_approval` marks a pending approval `rejected`. A global kill switch, `REMEDIATION_ENABLED`, makes `run_remediation_workflow` return early without consulting the LLM and makes `execute_approval` refuse to run, so the entire remediation path can be disabled with one setting.
- **Human surface.** The persistent `IncidentActionView` carries a Remediate button that runs `run_remediation_workflow` against the incident's stored diagnosis and posts one `RemediationApprovalView` (Approve/Reject, persistent) per proposal. Approve calls `execute_approval`; Reject calls `reject_approval`. The `/remediate`, `/approve`, and `/reject` commands (and their prefix equivalents) are fallbacks for when the buttons are not listening, such as after a bot redeploy.

The `pending_approvals` table schema is:

Table 3: SQLite `pending_approvals` table schema (module `agent.storage.repository`).

Column	Type	Notes
`id`	TEXT PK	8-character prefix of a UUID4
`incident_id`	TEXT	Incident the proposal belongs to
`container_name`	TEXT	Resolved container name
`container_id`	TEXT	Docker container id at proposal time
`action`	TEXT	e.g. `restart_container`
`status`	TEXT	`pending`, `approved`, `rejected`, `expired`, or `failed`
`created_at`	TEXT	ISO timestamp, used for TTL

4.8 Background monitoring

The background monitor, `agent.bot.monitor`, runs inside the bot process as a `discord.ext.tasks.loop`. The poll interval is read from the setting `monitor_interval_seconds` and floored at 10 seconds, so a misconfiguration cannot flood the network. Each cycle runs three checks.

4.8.1 HTTP endpoint liveness

For each `name=url` pair parsed from `MONITORED_ENDPOINTS`, the monitor calls `check_endpoint`. An endpoint counts as up when it is reachable and its status code is below 500. The previous state is kept in an in-memory dict. On any up-to-down or down-to-up transition, an info-severity alert is posted to the status channel. On a healthy-to-down transition only, a critical alert is posted to the alert channel and the incident workflow is run for that endpoint, opening a threaded incident.

4.8.2 PromQL alert rules

For each `name=<PromQL>` pair parsed from `PROMETHEUS_ALERT_RULES`, the monitor evaluates the query. A rule is firing when the result set is non-empty. Alerting is edge-triggered: a critical alert and a threaded incident are created only on the not-firing to firing transition, and a recovery note is posted on the firing to not-firing transition. If Prometheus is unreachable the rule is skipped for that cycle, but the endpoint liveness check still covers basic reachability.

4.8.3 Docker container monitoring

When `DOCKER_MONITOR` is enabled, the monitor lists containers each cycle:

- **Behaviour.** It alerts when a previously running container stops. New containers are picked up automatically with no configuration entry. A name filter (`DOCKER_NAME_FILTER`) narrows the watch set.
- **De-duplication state.** The de-duplication state is intentionally in memory, which is acceptable for a single-instance MVP. The trade-off is documented in the project goals: state resets on restart, which prevents duplicate alerts after a cold start but loses continuity across redeploys.

4.8.4 Periodic container-status snapshot

A second `discord.ext.tasks.loop` posts a compact status snapshot of the watched containers to the status channel on a fixed cadence (`STATUS_SNAPSHOT_INTERVAL_SECONDS`, default 900 seconds). The snapshot reuses the `_container_state` map that the container poll refreshes each cycle, so it adds no extra Docker query and is at most one poll interval stale. It is quiet

when no containers are watched or no status channel is configured, so an empty channel means "nothing watched" rather than "broken". This gives operators a standing heartbeat of container health in the same channel that records up/down transitions, without requiring them to run a command.

4.9 Incident tracking and storage

Incidents are persisted in a SQLite [12] database managed by `agent.storage.repository`, which holds two tables: `incidents` (this section) and `pending_approvals` (Section 4.7). The `incidents` schema is:

Table 4: SQLite `incidents` table schema (module `agent.storage.repository`).

Column	Type	Notes
`id`	TEXT PK	8-character prefix of a UUID4
`service`	TEXT	Service or endpoint name
`summary`	TEXT	LLM or fallback summary
`status`	TEXT	`open`, `acknowledged`, or `resolved`
`thread_id`	INTEGER	Discord thread id, if one was opened
`created_at`	TEXT	ISO timestamp

The repository resolves the database path relative to the repository root, so every entry point (bot, API, CLI, tests) shares one database. The store has a few properties worth noting:

- **Lightweight migration.** `init_db` adds the `thread_id` column if it is missing, so older MVP databases upgrade transparently.
- **Incident dataclass.** The `Incident` dataclass in `agent.storage.models` defaults `created_at` to the current UTC ISO timestamp and provides a `to_dict` helper.
- **Lifecycle.** A simple three-state machine: created as `open`, moved to `acknowledged` by `/ack`, and moved to `resolved` by `/resolve`, both of which post a follow-up into the incident's Discord thread if one exists.

4.10 Alerting

`agent.alerting.discord` posts rich embeds to Discord channels:

- **Severity embeds.** `send_alert` builds a `discord.Embed` with a title, a description, and a colour chosen by severity: blue for info, orange for warning, red for critical.
- **Incident threads.** `open_incident_thread` creates a public thread in the incidents channel, posts the summary into it, and records the thread id back into the SQLite row via `update_incident_thread_id`, so later ack/resolve updates can find the thread.
- **Persistent action views.** The thread message carries an `IncidentActionView` with Acknowledge, Resolve, and Remediate buttons; the view uses `timeout=None` so the buttons never expire while the bot is running, since incidents are often handled long after the thread opens. The Remediate button runs the gated remediation agent, which posts each proposal with its own `RemediationApprovalView` (Approve/Reject, likewise persistent). After a bot restart the buttons stop listening; `/ack`, `/resolve`, and `/remediate`/`/approve`/`/reject` remain as fallbacks.
- **Channel mapping.** The bot maps its output to six purpose-specific channels, all optional. A feature is simply inactive when its channel id is unset, so a deployment can adopt channels incrementally. The two highest-value channels for a demo are the alerts channel (so incidents surface) and the test-scenarios channel (to trigger reproducible runs).

4.11 Deployment and observability stack

The project ships a Docker Compose file that provisions the agent alongside a complete observability pipeline, so the whole system can be brought up with a single command. The stack contains:

- **api** and **bot**: the two agent services, built from a common `python:3.13-slim-bookworm` image using `uv` [13] with a cached dependency layer, both using host networking to reach the local Prometheus and Loki ports and the local LLM endpoint.
- **prometheus**: scrapes the nginx exporter and itself.
- **loki**: filesystem-backed log store.
- **promtail**: ships nginx access and error logs into Loki with a `service=nginx` label, so the agent's LogQL queries can select them.
- **nginx**: a sample monitored service exposing a `/nginx_status` stub-status endpoint, with real log files in a shared volume so Promtail can read them.
- **nginx-exporter**: the nginx metrics exporter feeding Prometheus.

Both agent services mount the host Docker socket read-only so that the container-discovery tools (`/status`, `/containers`, the menu) see host containers. The API service exposes port 8123 and has a healthcheck. The bot service is configured with `MONITORED_ENDPOINTS` pointing at the local API health endpoint and nginx, so the monitor has something real to watch out of the box.

4.11.1 Local LLM deployment and provider switching

The LLM is provider-portable without code changes:

- **Provider selector.** The `get_llm` factory reads a single `LLM_PROVIDER` selector and resolves it to a matching base URL and model from preset settings: `ollama` selects an Ollama endpoint, `vllm` selects a vLLM endpoint, and an empty value falls back to the legacy `LLM_BASE_URL`/`LLM_MODEL`. Local providers do not require an API key, so a dummy key is passed to `ChatOpenAI`, which mandates a non-empty one.
- **Separate vLLM stack.** The vLLM server runs as a separate Docker Compose stack outside the agent repository, keeping the GPU-bearing model server decoupled from the application compose. It serves an OpenAI-compatible endpoint on a host port and exposes the OpenAI tool-calling protocol through vLLM's `-enable-auto-tool-choice` flag with a model-appropriate `-tool-call-parser`.
- **Two model legs.** The Ollama leg serves a Gemma 4 model and tolerates GPU memory pressure through CPU offloading; the vLLM leg serves a text-only model sized to fit the available VRAM. Switching between the two is a one-line `LLM_PROVIDER` change followed by a bot restart.

4.12 Testing

The test suite, under `tests`, contains 81 test functions across 18 modules and is run with `pytest`. A shared fixture in `tests/conftest` points the repository at a temporary SQLite file, so storage tests are isolated. The suite covers five areas:

- **Workflows** (`tests.test_workflows`): the agentic loop with a fake LLM that returns scripted tool calls, the step-cap termination behaviour, the fallback path when the LLM

raises, the no-persistence property of the ask workflow, the use of the ranked-hypothesis prompt for incidents versus the general prompt for `/ask`, and the tool-calling probe.

- **Remediation** (`tests.test_remediation`): the pending-approval store, the structural gate (`propose_restart` records but never executes), `execute_approval` execution and TTL expiry, reject, the failure path, and the `REMEDIATION_ENABLED` kill switch.
- **Monitor** (`tests.test_monitor`): endpoint parsing, the healthy-to-down escalation state machine, edge-triggered rule firing, the container-status snapshot, and bearer-token header injection for both sources.
- **Tools and storage** (`tests.test_docker`, `tests.test_health`, `tests.test_ssh`, `tests.test_tools`, `tests.test_storage`): the Docker socket guard, HTTP health timing, systemd unit validation, the deterministic fallback-stub path for the source tools, and the full CRUD lifecycle of the incident repository.
- **Interface** (`tests.test_bot`, `tests.test_api`, `tests.test_alerting`, `tests.test_reporting`): command routing, API responses, embed colouring, and report formatting.

The suite uses no external services, so it runs in CI without provisioning Prometheus, Loki, or a live LLM. Test instances of the settings model are built with `_env_file=None` and, where needed, explicit field overrides, so that values from the deployment `.env` (loaded into `os.environ` at import time) do not leak into tests; the source-tool stub tests patch the underlying HTTP calls to raise, so the fallback path is exercised deterministically regardless of whether Prometheus or Loki is reachable. Tests marked `integration` exercise live endpoints and are opt-in.

5 Results and Impact

The MVP was delivered across the implementation phase of the internship and is functional against a live deployment. The measured outcomes are as follows.

5.1 Delivered capabilities

- **Integrated architecture.** The Discord bot, the FastAPI backend, the SQLite store, the shared LLM client, and the common tool set run as one codebase. A capability added to the tool layer is immediately reachable from both chat and HTTP.
- **Controlled agent tools.** Prometheus metric and label discovery, PromQL instant and range queries, Loki instant and range queries, HTTP health checks, Docker container discovery, and read-only Docker container status and logs tools are all implemented and exercised. The agent loop discovers real service names via `list_services` and matches them separator-agnostically. The incident lifecycle (open, acknowledge, resolve) is persisted. Docker and SSH mutation functions are library-only, reached through the gated remediation agent rather than bound to the investigator.
- **Gated remediation.** A second, human-gated agent proposes Docker restarts; each proposal is persisted, surfaced through Approve/Reject buttons or commands, and executed only on approval, with a time-to-live expiry and a global kill switch. The structural gate keeps the write path out of the autonomous loop.
- **Discord integration.** Prefix and slash command surfaces share one tool layer; the @mention conversational handler, the background monitor with its periodic container-status snapshot, the interactive container menu and service panel, the persistent incident action and remediation approval buttons, a reply-summarisation layer with a `raw` opt-out, and the six-channel output mapping are all delivered and tested.

- **Self-hosted, provider-switchable LLM.** The OpenAI-compatible client drives the agentic loop on any conformant server, with a single `LLM_PROVIDER` selector switching between local Ollama and vLLM backends, and a verified fallback when the configured model lacks tool-calling support.
- **Observability integration.** Prometheus and Loki sources return real data when reachable and degrade to stubs when not, with optional bearer-token authentication. A Docker Compose stack provisions the monitoring pipeline alongside the agent, and a separate stack provisions the vLLM model server.
- **Documentation.** The README, setup guide, usage guide, architecture overview, workflow reference, tool reference, and channel mapping are all maintained in the project's `docs` directory.

5.2 Quantitative scope

The implementation comprises approximately 2,871 lines of Python across 42 source files (excluding generated artefacts), 81 test functions across 18 test modules, and 56 commits over the implementation window. The Docker Compose stack orchestrates seven services (the agent's api and bot, Prometheus, Loki, promtail, nginx, and the nginx exporter), with the vLLM model server in a separate stack. The dependency surface is deliberately small: eight runtime packages.

5.3 Operational impact

In use, the agent reduces the time between a suspected degradation and a useful summary from a multi-tool, multi-dashboard manual process to a single chat command or a monitored escalation. The @mention path suits ad-hoc questions that do not warrant a formal incident, because it reuses the same agentic loop without persisting a record. The background monitor closes the gap for degradations that no one is watching, and the threaded incident gives the team a durable, searchable record per event. The investigation loop is read-only, so it can be left running in a real environment with confidence that it will not change infrastructure state; the remediation loop can change state, but only through a human approval, keeping the blast radius of any single action to one explicitly approved restart.

5.4 Verification

The test suite passes with no external services provisioned. That confirms the graceful-degradation design holds: the agent responds correctly when Prometheus, Loki, or the LLM is absent. The integration path is exercised manually against the Docker Compose stack, where the monitor detects the nginx service going down and opens a threaded incident with an LLM summary.

6 Challenges

Several challenges were encountered and resolved during the implementation:

1. **Tool-calling support varies by model.** Not every OpenAI-compatible server implements the tool-calling protocol. The `llm_supports_tool_calls` probe and the startup warning were added so an operator is told immediately that investigations will degrade to static summaries. The two-level fallback (agentic, then static LLM summary, then raw context) ensures the agent still produces output on an unsupported or offline model.
2. **Gateway reconnect re-entrancy.** The `discord.py` gateway can fire `on_ready` more than once after a reconnect. Without the `_setup_done` guard, the monitor and the log handler

would be started twice, duplicating alerts and pump tasks. The guard is a single module-level boolean, which is sufficient for a single-process bot.

3. **Discord message length.** LLM summaries and log dumps frequently exceed Discord's 2000-character limit. The `split_discord_text` helper splits at the nearest newline, keeping log lines intact. The code blocks are capped at 1900 characters to leave room for the backtick delimiters.

4. **Log forwarding feedback loops.** Forwarding logs to Discord risks a loop when the Discord client itself logs an error that triggers another log record. The `DiscordLogHandler` skips any record from a logger whose name starts with `discord` and writes send failures to `stderr` rather than the root logger, which closes the loop.

5. **Configuration normalisation.** Pydantic's `env_parse_none_str` converts empty environment variables to `None`, which broke string concatenation in several places. The `@model_validator(mode="before")` in the settings class normalises `None` back to `""` for string fields, and this fix is covered by a test.

6. **In-memory monitoring state.** Storing alert de-duplication state in memory is the simplest correct choice for a single-instance MVP, but it means state resets on restart. Persisting this state is deferred to future work, and the trade-off is documented.

7. **Docker socket availability.** The Docker tools must not crash when the socket is absent (as in a non-Docker deployment or a container without the socket mounted). The `_docker_available` guard and the try/except around the API call make the tools degrade to empty results, which keeps the bot responsive in any environment.

8. **Multimodal model footprints on a constrained GPU.** A local vLLM server must fit the model and the KV cache in VRAM, which it cannot offload. The Gemma 4 instruct checkpoints are unified multimodal bundles (text, vision, and audio towers), so even the size labels that imply a small text head load all towers and exceed an 8 GiB card. Ollama tolerates the same models through CPU offloading, so the Gemma 4 leg runs under Ollama; the vLLM leg runs a text-only model sized for the available VRAM. The two legs are exposed through the `LLM_PROVIDER` selector so an operator compares backends without touching code.

9. **Tool-calling probe fidelity.** A startup probe that asks the model to "pick a tool" underreports capable local models, which answer such a vague prompt in prose while still calling tools on real investigations. The probe issues a concrete task that requires a tool (listing containers through `list_services`), so a capable model emits a tool call and the probe reports support correctly.

7 Future Work

The MVP is a foundation for several planned extensions, listed in priority order.

1. **Additional gated runbooks.** Extend the remediation agent beyond container restart to the other high-value, low-risk on-prem runbooks (clear disk space, rotate logs, restart a Docker daemon), each reusing the existing propose-then-approve gate and kill switch.

2. **Promote runbooks to autonomy individually.** Graduate a runbook from human-approved to autonomous execution only after a track record of correct approvals, using the history already captured in the `pending_approvals` table; each promotion keeps an allowlist entry, a blast-radius limit, and the kill switch.

3. **Real service inventory for status.** Wire `get_service_status` to a fuller registry (Docker labels, a service catalogue, or an orchestrator API) so status reporting reflects more than container liveness.
4. **Named multi-environment sources.** Add a registry of named monitoring sources (production, staging) so the agent can be asked about a specific environment rather than a single global Prometheus/Loki pair.
5. **Persisted monitoring state.** Move the alert de-duplication state into SQLite so it survives restarts and redeploys without duplicate alerts.
6. **Persistent action buttons across restarts.** Re-register the incident action and remediation approval views on startup from the open incidents and pending approvals in SQLite, so the buttons keep working after a bot redeploy instead of falling back to the `/ack`, `/resolve`, `/approve`, and `/reject` commands.
7. **Role-based access control.** Restrict sensitive commands and tools to authorised roles, and add an allowlist for SSH and Docker actions.
8. **Alertmanager webhook receiver.** Host a webhook receiver in the bot process as an alternative to in-bot polling, letting an external Alertmanager push alerts rather than having the bot poll Prometheus.
9. **Structured evaluation of the agentic loop.** Build a scenario suite that scores the agent's ranked hypotheses against expected findings, so model and prompt changes can be measured rather than judged by hand.

8 Conclusion

The Marconi Infrastructure Operations Agent demonstrates that a useful, safe, and self-hosted operations assistant can be built as a small Python codebase that reuses a mature LLM tool-calling protocol over read-only infrastructure tools. The deliberate read-only scoping lets the agent run in a real environment, the graceful-degradation design keeps it responsive during partial outages, and the shared workflow layer means the Discord and HTTP surfaces never drift. A gated remediation agent introduces the first write action behind a human approval, a time-to-live expiry, and a kill switch, so the autonomous investigation loop gains a tightly scoped, reviewable remediation counterpart. The MVP meets the internship charter and leaves a clear, prioritised path toward more runbooks and, eventually, individually promoted autonomous remediation.

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A Discord channel mapping

The bot routes its output to six purpose-specific Discord channels, each opt-in via an environment variable. A feature is inactive when its channel id is unset, so a deployment can adopt channels incrementally.

Table 5: Discord channels and their roles.

Channel	Env var	Role
`#agent-commands`	(none)	Operators run `!` commands; the bot replies inline
`#alerts`	`DISCORD_ALERT_CHANNEL_ID`	Severity-coloured embeds for down-alerts and incidents
`#status-updates`	`DISCORD_STATUS_CHANNEL_ID`	Every up/down transition the monitor observes, plus the periodic container-status snapshot
`#incidents`	`DISCORD_INCIDENT_CHANNEL_ID`	One thread per incident with ack/resolve/remediate updates and approval views
`#agent-logs`	`DISCORD_LOG_CHANNEL_ID`	WARNING and above bot and back-end log records
`#test-scenarios`	`DISCORD_TEST_SCENARIOS_CHANNEL_ID`	`/scenario` runs, restricted to this channel when set

B Scenario file format

Scenarios are YAML files in the `scenarios` directory. The `/scenario <name>` command loads `scenarios/<name>.yml` and runs the incident workflow for the named service. The sample scenario, `scenarios/sample_slow_query.yml`, models a slow query on PostgreSQL:

Listing 2: Scenario file format (YAML).

```yaml
name: Slow query on PostgreSQL
service: postgres
symptom: >
  A query plan change causes CPU to spike and response times to climb.
expected_agent_actions:
  - retrieve service status
  - query Prometheus for cpu usage
  - query Loki for slow query logs
  - summarize findings
  - store incident
  - alert Discord channel if configured
```

C Module reference

The table below lists every implementation module by its fully-qualified import path, the canonical way to locate code in this project.

Table 6: Module reference.

Module	Responsibility
`agent.cli`	Command-line entry points for bot, API, incident, report
`agent.config.settings`	Pydantic settings loaded from env and `.env`
`agent.config.dotenv_loader`	Explicit `.env` loading before settings
`agent.bot.client`	Discord client, on_ready, on_message, log handler
`agent.bot.commands`	Prefix command handlers
`agent.bot.slash`	Slash command handlers (CommandTree)
`agent.bot.menu`	Interactive container picker and action buttons
`agent.bot.views`	Service panel view and persistent incident action buttons
`agent.bot.summarize`	LLM reply summarisation with `raw` opt-out
`agent.bot.monitor`	Background endpoint, rule, and container monitor
`agent.bot.utils`	Discord text splitting and Prometheus/Loki formatting
`agent.api.server`	FastAPI app and shared-secret dependency
`agent.api.routes`	HTTP API routes
`agent.workflows.incident`	Agentic investigation loop, ask workflow, and probe
`agent.workflows.remediation`	Gated remediation agent, approval store, kill switch
`agent.workflows.reporting`	Incident report formatting
`agent.tools.services`	Service status via Docker container discovery
`agent.tools.metrics`	Service metrics via Prometheus with escape and fallback
`agent.tools.logs`	Service logs via Loki with fallback
`agent.tools.health`	HTTP endpoint health checks
`agent.tools.docker`	Docker Engine API tools; mutation functions are library-only, reached via the gated remediation agent
`agent.tools.ssh`	SSH-based server tools with unit-name validation
`agent.tools.normalize`	Separator-agnostic service-name matching
`agent.tools.incidents`	Incident action tools over SQLite
`agent.tools.query_tools`	Read-only LangChain tools the LLM calls
`agent.tools.sources.prometheus`	Prometheus query, discovery, and range helpers
`agent.tools.sources.loki`	Loki instant and range query helpers
`agent.storage.models`	Incident dataclass
`agent.storage.repository`	SQLite incident and pending-approval repository
`agent.alerting.discord`	Discord alert embeds, incident threads, and action views
`agent.llm.client`	OpenAI-compatible ChatOpenAI factory with provider switching